The Electrons Must Flow

What I learned mapping companies from the turbine to the token.

> *"The spice must flow." — FRANK HERBERT, DUNE*

Frank Herbert's *Dune* is remembered for its deserts, its religion, and its politics, but underneath all three it is a book about an economy that runs on a single flow. Herbert imagined a universe in which travel, trade, and power all depended on one substance from one planet, then followed what that dependence does to prices and politics. His answer fits in four words, repeated like a law of nature: “the spice must flow”. Everyone says it because everyone's power depends on it. The logic underneath the four words transfers exactly to the energy-to-compute market. When everything depends on a flow, power belongs to whoever stands where it narrows.

The flow I watch runs from electricity to intelligence, and the clearest way to see what a position is worth is to look at what people pay for time on it.

Let’s look at two contracts which are a few months apart. Anthropic signed a twenty-year lease worth roughly nineteen billion dollars in contracted revenue for a Kentucky campus that does not exist yet: a patient buyer, planning ahead. Reflection agreed to pay SpaceX $150 million a month for immediate access to ready GPUs at Colossus 2 in Memphis: an impatient buyer, paying now. Work both out per unit of energy, as the power analyst Neel Somani brilliantly did, and the patient buyer pays about $271 per megawatt-hour while the impatient one pays the equivalent of nearly $5,000.

The comparison is not clean, and Somani says so himself: one price includes racks of ready GPUs, the other only the building and the cooling around them, which is why he calls his own number napkin math. But subtract the hardware and a large premium survives, and it prices the same thing. Not electrons. When. In this market the hardware is written off with barely a shrug, reactors abandoned mid-permit, GPU fleets depreciated in three years, but nobody writes off time. Every dollar in this essay is, one way or another, a price paid for time.

I wanted to understand who gets paid in a market that prices time like that. The problem is that the question runs through half a dozen fields at once: siting and permitting, grid engineering, nuclear licensing, power electronics, thermodynamics, chip packaging, market structure all to name a few. Nobody is an expert in all of them. I'm certainly not. So I did the thing I could do. I broke the flow from electricity to intelligence into eleven stages, from siting and permits at one end to compute orchestration at the other, and mapped the early-stage companies forming at each, roughly 450 of them. You will not find public and late-stage companies as they already have filings, analysts, and endless discourse covering them. This map is of the private layer, where founders are still choosing stages and where there are opportunities. The set skews North American, with some European coverage and less elsewhere, and it is certainly not complete. It doesn't need to be. The point was never the list, it was to see what patterns emerge when you sort the buildout by physics instead of by market category and then count, and a base this size is enough for the patterns to show.

What is the buildout, physically?

Strip away the logos and it is a single process. Somewhere in west Texas, at this moment, an electron is leaving a turbine. It departs at 18,000 volts and is pushed up to 345,000, because high voltage is how electricity travels far without losses. It crosses a hundred miles of transmission line, drops through a substation, and threads a transformer the size of a school bus that took three to five years to procure. It then enters a building that did not exist two years ago. Inside, it falls to 800 volts in the newest halls, then to 48, then below one, descending conversion after conversion like a river descending locks, and each lock spills something, one to a few percent per stage, five or more stages deep. It arrives at a processor no wider than a playing card and exists there, as useful energy, for a few nanoseconds. In that instant it flips transistors that multiply enormous matrices of numbers, and those multiplications, stacked layer on layer, become a prediction of the next word of someone's question. Then the electron is gone. A token leaves the building as light down a fiber, and somewhere far away, a sentence grows one word longer.

(A physicist will object that no single electron makes this trip, and the physicist is right. Electrons in a wire drift at a walking pace measured in millimeters per second, and what actually travels, at a good fraction of the speed of light, is the energy in the field around the wire. That said the fiction is worth the clarity, so I will keep it.)

Two facts about this flow set the terms for everything else.

- First, it is lossy. Between the grid and the die, power is converted five or more times, each conversion shaving one to a few percent. NVIDIA's own 800-volt materials put today's end-to-end electrical chain below 90 percent efficient, with industry estimates running from the low eighties up. Then the heat has to be hauled back out, which at a well-run site (a power usage effectiveness of 1.2 to 1.3) takes another 17 to 23 percent of everything entering the fence. Put the two together and only about three quarters to four fifths of the power at the fence ever reaches the IT equipment, which then converts it again on board before any of it becomes a token.
- Second, the flow is only as fast as its narrowest point. It does not matter how many GPUs you own if your grid connection is in year three of study, similarly it does not matter how fast your chip is if it is waiting on memory. The flow has one throughput, set by whichever stage is currently the bottleneck, and the bottleneck moves like the worm beneath Herbert's sand: you feel where it surfaces, and you listen for where it will surface next. Three years ago it was chips. Then transformers. Then interconnection. Then cooling. Then memory. Right now, in different places, it is all of them at once.

That is the physics. Here is the frame it forced, which I ended up testing everything against: **in a lossy flow with moving bottlenecks, the interesting opportunities are the ones that either shorten the flow or own a point it cannot route around.**

What are the two moves, exactly?

The two moves deserve careful definitions, because the rest of the essay leans on them. And the first thing to notice is that they are not the same kind of claim. Shortening is a bet about physics. Owning is a bet about power. The whole game, as we will see, is turning the first into the second.

A company **shortens** the flow if, holding everything else constant, its existence means more tokens come out per joule, per dollar, or per month of waiting. The test is physical, delete the company from the world and ask whether the flow gets longer, slower, or leakier. Shortening comes in three forms.

- Remove losses: recover part of the energy that never reaches the chip. This is the stillsuit's move, reclaiming the water in your own breath, because on Arrakis every loss is fatal.
- Remove steps: delete a conversion, a study, a manual process entirely.
- Remove waiting: compress a queue, a bring-up, a permit.

All three count, but they are not priced equally. Recover a loss and you save percentage points of energy cost. Delete a step and you save its loss plus its equipment queue. Delete waiting and you move the project's entire revenue forward in time. That is the largest effect of the three, because it applies to the whole asset rather than to one line item. The spread at the top of this essay, the same megawatt-hour selling for $271 to a patient buyer and nearly $5,000 to an impatient one, is the market putting a number on exactly this.

And be clear about what fails the test. A marketplace renting you the same GPU more cheaply has moved money between pockets, not shortened anything. Software that produces a prettier permit application but no earlier approval date has shortened nothing. A consultancy that helps you wait more knowledgeably is selling commentary on the queue, not a shorter queue.

A company **owns a chokepoint** if the flow cannot route around it, and the value of the position follows one rule: a chokepoint charges what the detour costs. If every alternative path costs a competitor two years, the tollbooth on the short path can charge up to two years of value, and the traffic pays. At a chokepoint, the position is the moat.

But not everything that looks like a chokepoint is one, and the difference is what the position is made of. Real chokepoints are made of physics: high-bandwidth memory that only three companies on earth can fabricate, because the constraint is packaging rather than design; enriched fuel that requires a licensed facility; energized land whose grid connection took five years to win. Or they are made of law: an exclusive license, an approved design, a seat in a regulated market (Herbert knew this one too. All Imperial commerce routed through one company, CHOAM, and the real prize in his universe was a directorship). They can also be made of institutional processes or highly specialized talent: a fab's yield learning, a utility's vendor qualification, knowledge that lives in feedback loops rather than in any one person's head. So when someone claims a moat, ask three questions: what is it made of, what is its half-life, and what is it, in turn, hostage to? A transformer queue is a chokepoint today and is being competed away as we speak. Enriched fuel and HBM are structurally harder to route around. And even the hardest positions hang from something upstream: HBM is hostage to advanced-packaging capacity and to new DRAM fabs, and a new DRAM fab is hostage to construction crews and a grid connection, which is where this essay began. The detour always gets cheaper eventually, the question is whether it takes two years or twenty.

One more thing, and it took me the longest to see. The two moves are less alternatives than a sequence, and the sequence is how a physics bet becomes a power position. Shortening creates value but struggles to keep it, because a shortener sells into a competitive market for the thing it shortens. The time it saves is mostly captured downstream, by whoever owns the asset that starts earning sooner. Owning captures value but rarely offers an opening, because the good chokepoints are mostly claimed. So the companies I find most interesting run the sequence, shorten to enter, and let the shortening accumulate into something ownable such as a dataset, a delivery slot, a qualified-vendor status, an installed control plane. Ask every shortener what the saved time compounds into. If the answer is a stronger sales pitch, it is a supplier having a good quarter. If the answer is a position, it is a chokepoint in formation.

Where are the founders actually showing up?

I expected company formation to be spread along the flow. It isn't, the shape is a dumbbell. But be careful which distribution you look at, because two matter here and they say different things. The raw count of companies per stage measures accumulated attention: generation is the biggest stage on the map at 96 companies, orchestration holds 79, compute silicon 58. The share of each stage founded recently measures something else, where pain arrived lately, and that is the distribution shaped like a dumbbell.

In siting and permits, 54 percent of the stage's 41 companies were founded in 2024 or later. In compute orchestration, 54 percent of its 79. Through the middle of the flow the share runs between the teens and low thirties, bottoming at 12 percent in the construction stage, where only 4 of 33 companies are that young. Even the bottom is informative rather than strange: construction tech's founding wave came a decade ago, for other customers, and that older cohort simply added data centers to its market. The low share marks a stage served by pre-existing companies, not an unserved one.

And the shape is not an artifact of how the stages are sized or grouped. The cleanest way to see it needs no pooling at all: the two end stages hold 27 percent of the map's companies but attracted 46 percent of everything founded since 2024, which works out to about 32 new companies landing at each end stage against 8 at each middle stage, roughly four to one. Drop generation, the largest stage, from the middle entirely and the middle's rate does not move. The one bias I cannot compute away is my own, since this is a map I built rather than a census but the pattern survives every check I can run on it. Two checks I cannot fully run deserve naming anyway. Software companies are simply faster to found than hardware companies, so some of the dumbbell may be formation speed rather than pain. And a 2024 startup with a launch post is easier to find than a 2019 test lab in Ohio, so some of it may be my search process. Both are real, and I cannot fully separate them from the pain story with this data. What neither explains is the middle's particular combination of old incumbents and lengthening physical backlogs, which is not a founding-rate artifact of any kind. (All vintage figures exclude a few entries with no founding date, which is why a count here occasionally reads one lower than the terminal's.)

When a stage shows old incumbents, no recent entrants, and a lengthening backlog, the absence of founders is not evidence of health. It is Herbert's worm again: the sand is quiet, and no one is listening for what moves beneath it. And that is what this distribution is actually for. If founding waves mark where pain arrived, the question that pays is where the wave goes next, and combined with the two patterns below, closing windows and published specs, the map starts generating predictions. Some are already coming true as stops on the walk, answered by companies this essay features.

Others are still empty columns. Switch-level optics shows the full signature of an aging cohort graduated to unicorn scale, a co-packaged-optics roadmap printing now, almost no young entrants. Water as procurement rather than cooling, the rights, reuse, and acquisition layer that already decides permits in Texas, has no seats taken at all. Nuclear's operational layer, the licensing throughput, fleet software, and qualified supply chain a family of small reactors will need, has exactly one early arrival. And those are just a few of them, the map holds more.

What happens when a window closes?

Stage by stage the same life cycle repeats, and it is worth slowing down on the mechanics because this pattern does the most work in the walk. A bottleneck bites somewhere on the flow. A cohort of startups forms to attack it. Capital floods in. One of them wins, and the winner consolidates the stage, it gets acquired at a scale no seed check can follow, or it raises at a valuation that already prices the problem as solved. That is what I mean by a window closing. The chance to back a small company against that specific problem is gone, because the problem is no longer open, you would be paying billions for certainty that was available at seed prices three years earlier.

But here is the part that makes closed windows useful rather than just sad. The winner actually fixes the stage, and once a stage is fixed it stops being the thing that limits the flow. The limit reappears at the next weakest stage beside it, where there is no winner yet and the companies are still small. Cooling ran the full cycle in under four years: GPUs got too hot for air, a cohort formed, direct-to-chip liquid cooling won, and Ecolab bought CoolIT for $4.75 billion, the largest pure-play data-center cooling deal ever (Eaton paid nearly twice that for Boyd months earlier, but Boyd is a broad thermal business; CoolIT was all data center, which is the point). That closed the chip-level window, and it moved the problem rather than ending it, liquid loops unlocked denser racks, the heat still has to leave the site, and the constraint now sits at the facility boundary, where a new cohort is forming at seed prices.

The map keeps more receipts like that, and each one reads the same way, what is closed, and where it points. RadixArk raised $100 million at a $400 million valuation within a year of founding, for software that squeezes more inference from existing memory: the software route through the memory wall is priced as solved, so the open doors are the physical ones, interconnect, optics, packaging. Baseten serves models at a reported $600 million annualized revenue: the platform layer is built, which points down, at the primitives still missing beneath it, like moving a running job. Etched at $10.3 billion for its transformer-only chip: specialized silicon has graduated, which points sideways, at the software on-ramps every new chip still lacks. Armada raised $230 million at a roughly $2 billion valuation for modular data centers: that category has its winner, and the open questions moved inside the module, to the power and cooling systems it is assembled from.

General Matter holds a $900 million Department of Energy contract to restart domestic uranium enrichment: the fuel chokepoint is claimed, which points up the stack, at the reactor cohort that must now buy from it. Aalo reached criticality (a self-sustaining chain reaction, the moment a reactor first turns on) on its first reactor: the reactor cohort is graduating on schedule, which points at the enablers, licensing, siting, operations software, arriving on the same calendar. Compute Desk publishes live compute price indices carried on 350,000 Bloomberg terminals and is building the deal, escrow, and settlement layer around them: the reference-price seat in compute is being taken as I write, which points beneath it, at the physical layer, because paper only trades as well as the underlying compute can actually be delivered and moved.

In every case the interesting question is not who won the closed stage. It is where the constraint went, because that is where the next cohort is forming while it is still cheap.

Who writes the industry's calendar?

The industry announces its own future in writing, years ahead, and then finances the companies that must exist to meet it. When a dominant player publishes a new spec, it simultaneously publishes a list of components that do not exist yet, and a dated cohort of companies forms within quarters to build them. NVIDIA's 800-volt DC architecture is the clearest case: the published roadmap named the power-electronics parts the transition requires, wide-bandgap devices, solid-state transformers, 800-volt-to-sub-one-volt converters, and a cohort formed to build exactly those. The co-packaged-optics roadmap is doing the same thing now for photonic interconnect. And the spec-setter does not just publish the calendar, it finances it. By my count, NVIDIA sits on 18 cap tables across eight of my eleven stages, with dozens more companies carrying its imprint through Inception and reference designs. When the spec-setter tells you what must exist next, believe it. Two updates keep this honest. The calendar is no longer one company's: the 800-volt work is now published jointly with Google and Microsoft through the Open Compute Project, with dozens of vendors signed on, which makes the dates more binding, not less. And there is a bear case worth stating plainly. If the spec-setter publishes the calendar, finances the cohort, and controls the reference design, some of these companies are less startups than a supplier-development program wearing venture pricing, with exits capped at strategic-acquisition prices. The calendar de-risks timing. It does not promise the value lands with the cohort.

And the lens underneath all of this, carried down the whole flow as a question: which parts of the accepted path are physics, and which are history? Solar panels produce direct current. Batteries store direct current. Chips consume direct current. Everything between those endpoints, the inversions and step-ups and step-downs and rectifications, is the standard path, and it was built for a different problem: moving AC power long distances to spinning motors. History can be deleted. Physics cannot. Herbert drew the same line: his Empire spent ten thousand years failing to synthesize the spice, and that failure is the only reason Arrakis mattered.

Now let’s take these lenses and walk. Eight stops, each framed the way I actually encountered it: as a question.

Stop one: can software release megawatts that already exist?

The flow does not begin with power. It begins with information, twice over, and both times the information is wrong or missing. First, siting. Which parcels can host a gigawatt campus is knowable from existing records, substation loading, feeder capacity, fiber, water, zoning, but the industry assembles that picture through consultant engagements measured in months per site. At the top of the flow, every month compounds through everything downstream.

Second, headroom. Utilities study new data centers as static, worst-case loads that can never flex, when a modern AI facility can dial its draw down and reshape it in milliseconds. Duke's Nicholas Institute with lead author Tyler Norris quantified what that mismatch conceals in [Rethinking Load Growth]: 76 to 126 gigawatts of new load could be absorbed on the existing grid across the 22 largest balancing authorities that are the regional operators that keep supply and demand matched. The condition is that large customers accept curtailment (being briefly turned down when the grid is tight) a fraction of a percent of the time. 76 gigawatts of headroom at 0.25 percent, 126 at 1 percent. For scale, the upper end is more than a hundred gigawatt-class power plants' worth of headroom, available without pouring any concrete. The authors are careful about the limits and I should be too: the study does not model transmission constraints, and the headroom is a one-time bridge measured against historical peaks, not a renewable resource. The megawatts exist. The description of them is wrong, and correcting a description is software work.

Three companies do that work from three positions, and the positions matter as much as the products.

Build works from the developer's side of the wall. It calls itself an agentic development firm, which in practice means AI agents paired with human development experts running the workflows that precede any construction, site sourcing against a buyer's criteria, desktop diligence, permitting submissions formatted to each authority's requirements, feasibility and underwriting. Underneath the agents sits the asset that makes them work, a geospatial engine mapping power capacity, substation loading, fiber routes, environmental risk, and parcel-level zoning across eighteen-plus countries, updated continuously and cited in every output. The traction is unusual for a seed-stage company because the product replaces consultant engagements that developers already pay for. 250-plus projects delivered across seventeen countries, for clients including Tishman Speyer (which liked the work enough to invest through its venture arm), STACK Infrastructure, and the UK government's program to identify national AI sites. Run the test on it: delete Build and months of waiting return to every project it touches. That is the shortening move. And it compounds into something ownable, because every engagement deepens the parcel-level corpus that any later entrant would have to buy or rebuild.

Senpilot works from inside the utility, which is the side of the wall software rarely reaches. The years a data center spends waiting are mostly spent inside the utility's own processes, interconnection studies, capital plans, rate cases, regulatory filings, all of it produced manually by engineering teams that are chronically understaffed for a load-growth era they never planned for. Senpilot builds AI tooling for modern utilities across exactly those workflows, for grids straining under electrification and data-center load at once. It is a small company attacking the least glamorous bottleneck in the entire flow, and that is the point, very few are standing where it stands, and its headcount has roughly tripled in a year because utilities, the slowest buyers in the economy, are buying. The thesis connection runs through a fact about queues: interconnection years are the largest single block of waiting on the whole flow, and they cannot be shortened from outside, because the queue lives inside the utility's own processes. Faster studies and faster filings are the only lever that reaches that waiting, and Senpilot is building the lever.

Soma Energy works across the boundary, on live assets rather than paperwork. The founding team scaled AWS's renewable energy program to roughly ten gigawatts, which means they spent years inside the largest corporate energy buyer on earth watching where flexibility goes unused. Now independent, they run an AI platform on both sides of the market at once. For power producers, that means autonomous trading and dispatch, about two gigawatts of generation and storage so far. Knowing when to sell, store, or dispatch makes the same assets earn more and waste less, which is more sellable power without any new steel. For data centers, it means faster time to power and more capacity from the sites they already have, with procurement, batteries, and workloads orchestrated together. Sitting on both sides is the product. The platform that sees how producers dispatch and how loads can flex is the one positioned to match the two. Every megawatt of flexibility they can prove and orchestrate is a megawatt of headroom the grid can serve without building anything, which is the Duke arithmetic converted from a white paper into a P&L. The shortening here is deleted waiting at the largest scale available: a flexible load gets served by the grid that already exists instead of queueing for the grid that must be built. And the ownership is the installed base itself, live assets under management on both sides of the meter. A pitch deck can be copied, two gigawatts under orchestration cannot.

Stop two: is nuclear a technology bet or a paperwork bet?

If the grid cannot serve you fast enough, generate on your own land. The honest ordering here is that gas is the three-year answer and nuclear is the thirty-year one. What is physically being installed behind meters for 2026 through 2029 is gas turbines, engines, and fuel cells; the turbine makers' backlogs are sold out toward 2030 and slot reservations trade like assets, and my map holds a whole cohort of behind-the-meter gas companies I am not walking here. I walk nuclear instead for two reasons: it is the generation everyone wants to own for the next thirty years, and its constraint structure teaches the two moves more cleanly than any other stage. Because here the flow narrows somewhere unusual. Fission's physics has been settled for fifty years; what gates deployment are documents: an approved design, licensable fuel, enrichment capacity. Watch what gets bought. General Matter holds one of three $900 million DOE awards to restart domestic enrichment of HALEU (the higher-enriched uranium fuel advanced reactors run on) at Paducah: the fuel chokepoint is being claimed, though the claim is a milestone contract and a plan, not yet an operating plant. And Aalo Atomics brought a full-scale test reactor to first criticality at Idaho National Laboratory on July 4, 2026, built in about eight months, the zero-power physics step that precedes its sodium-cooled power reactor rather than the power reactor itself, but a real chain reaction achieved on a startup's schedule. The cohort is keeping its calendar.

Inside the reactor cohort, the two moves appear in their purest forms, one company each.

Apollo Atomics is running the shorten move against two slow clocks in American infrastructure at once: the Nuclear Regulatory Commission's docket and the construction site. Its founder, an MIT nuclear PhD, asked one question: what single design change most collapses nuclear's cost and build time? The answer is the steam generator, the plant's largest component. Apollo shrinks the channels inside it, which packs more heat-transfer surface into less volume the way shrinking transistors pack more compute onto a chip, at a claimed factor of twenty. Everything else stays ordinary on purpose, standard low-enriched fuel and light-water physics the NRC has approved for seventy years, so the regulator evaluates almost nothing new; pre-application is underway, with a construction permit targeted for 2028. Shrink the largest component and the reactor, the building, and the schedule all shrink with it: factory-assembled, truck-delivered in claimed classes up to 300 megawatts, installed in under 24 months against the industry's decade. And since over half a first plant's lifetime cost is capital paid out during construction, halving the build time attacks nuclear's cost where it actually lives.

Applied Atomics took the other move, and wrapped a delivery business around it. The company builds, owns, and operates on-site nuclear plants from 100 megawatts up to a gigawatt, selling customers firm power behind their own fence rather than selling them a reactor, which is the shape of product the data-center buildout actually wants to buy. What makes that model possible without a decade of design work is the position underneath, an exclusive land-based license to BWXT's mPower, a small modular reactor one of the country's most credentialed nuclear manufacturers had already engineered, with BWXT retaining the IP and manufacturing role. One honest clause the press release skips: mPower is a program its original owners shelved in 2014 after hundreds of millions of dollars, and no design certification ever issued, so the license buys a decade of engineering head start, not an approved design; the certification work restarts from cold. The license is still the ownership move, a document competitors cannot route through however the cohort resolves; the build-and-operate model is what converts the document into revenue. Reactor startups are usually pitched as technology bets. Sorted by the flow, these two are constraint bets, and the constraints are made of exactly the materials that hold: physics and law.

So: a paperwork bet, at the margin. The sharpest counterargument says nuclear's real constraint is industrial capacity, not documents: qualifying a new fuel takes years of irradiation no regulatory reform can compress, and nuclear-qualified welders compete with data-center construction for the same hands. Both are true at different horizons. Paperwork gates the next handful of deployments; industrial capacity gates the next hundred. This stop is about the next handful, and it is why nuclear is the one stop where owning beats shortening outright: Apollo can compress the clocks, but Applied's license and General Matter's fuel position are things no competitor can route around without a decade of NRC review.

Stop three: why does a transformer take four years, and who captures the time you save?

Here the physics-or-history lens cashes out. A data center with generation on its own fence needs neither the distance nor the motors the AC grid was designed for, which makes most of the conversion chain history rather than physics. Voxel Energy, founded by ex-Tesla and EV-conversion engineers, takes the argument to its conclusion: rapidly deployed data centers with their own solar-plus-battery generation, run as an all-DC system from the highest voltage at the panel down to the lowest at the load, with battery packs sized so that training-workload surges that would destabilize an AC grid barely register. The result is two conversion stages at as little as 2 percent loss each, roughly 4 percent total. The same solar-and-battery system assembled from conventional AC-grid components would chain six or more stages and, by their white paper's arithmetic, lose as much as 30 percent. Whatever wins the category, the arithmetic is the point, deleting a stage removes its loss and its queue at once. Voxel is the one company on the map running all three forms of the move at once: removed losses (up to twenty-six points, by their own arithmetic), removed steps (the stages themselves), and removed waiting (no interconnection queue, because there is no interconnection).

A lot of the loads will stay on the grid, though, and the grid's binding queue is a machine whose design is a century and a half old. Transformer lead times reached four years, and not because transformers are hard to make. American electricity demand was essentially flat for nearly two decades, so every manufacturer that bet on growth got burned, and the survivors sized their factories for replacement demand at the exact moment demand began compounding. The queue is institutional memory, cast in factory capacity, disbelieving the future in public. Ayr Energy arbitrages the disbelief: substation- and transmission-class transformers through Indian contract manufacturing, where capacity never stopped growing, delivered in about a year against the industry's three to five, with a $500-million-plus order book covering more than twenty gigawatts.

Now run the naive arithmetic on that order book, because it answers the second half of this stop's question and teaches the sequence better than any example I have. Five hundred million dollars across twenty gigawatts is about twenty-five dollars a kilowatt, which is roughly what transformers cost. Ayr hands its customers two years and charges them transformer prices. The time value it creates is captured downstream, by the developer whose project starts earning sooner. That is not a flaw in the strategy, it is the general condition of shortening. And it sharpens the half-life question, because this particular chokepoint is being competed away in real time. GE Vernova's electrification segment booked $2.4 billion of data-center equipment orders in the first quarter of 2026 alone, more than in all of the prior year. Ayr's durable prize is not the toll. It is qualified-vendor status: a place on utility approved lists that takes years of testing and field acceptance to earn, and that competitors cannot hire their way onto. Shorten to enter. Own to stay. The honest caveat is that the second half is still a bet, and a hard one: utility qualification tends to freeze the exact manufacturing flexibility Ayr is selling, and if the order book skews toward developers and independent producers, who keep no approved lists, then the company has so far demonstrated only the first half of its own sequence. Still the essay's cleanest specimen, the whole idea priced in a single order book at twenty-five dollars a kilowatt.

Stop four: what must exist for NVIDIA's roadmap to happen?

Once inside the fence, the electron's remaining journey is conversion after conversion, and this is where the spec-calendar pattern turns into specific companies. NVIDIA's published 800-volt roadmap collapses the data center's four-plus conversion stages to two, lifts end-to-end efficiency by a sizable margin, and cuts copper use nearly in half. Note that this is a roadmap, megawatt racks targeted for 2027, not shipped reality, which is exactly what makes it a calendar. The next page is already printed: backside power delivery, where power enters the transistor from underneath the die, with Intel's PowerVia shipping now and TSMC's A16 targeted for volume production next year. Read the pages in order and they describe one continuous migration: conversion moving stage by stage from the substation toward the silicon, each step announced years ahead, each step dating a cohort of companies that must now exist. Every step of that migration is a shortening, deleting stages, losses, and copper at once.

Vertical Semiconductor is what the pattern predicts, arriving on the pattern's schedule. It spun out of MIT's Palacios Group, one of the world's leading gallium-nitride research labs, to commercialize a device the 800-volt world suddenly needs: GaN transistors built vertically, with current flowing down through the material rather than across its surface. The orientation is the product. Lateral GaN, the kind in laptop chargers, tops out at modest voltages and spreads across expensive die areas; a vertical device withstands higher voltage in a smaller footprint, which is precisely what converting an 800-volt bus down toward a one-volt chip inside a crowded rack requires. The company claims up to 30 percent efficiency gains in half the power footprint, raised an $11 million seed led by Playground Global, and counts Shin-Etsu, the Japanese materials giant whose substrates the process depends on, as a strategic investor, which for a semiconductor startup is supply-chain positioning as much as capital. The honest caveat is also the bet, vertical GaN is still largely pre-commercial because bulk gallium-nitride substrates are expensive, which is why the substrate supplier is on the cap table. And the bet has a graveyard, which is fairer to name than to hide: the last two American vertical-GaN pure-plays both died within the past three years, one bankrupt with its fab sold to onsemi, one wound down, largely on exactly those substrate economics. The pattern predicts the cohort; it does not protect its members. Vertical Semiconductor sits exactly on the seam the migration exposes between the building's new spine and the package. And if the bet lands, the position is made of the materials this essay trusts: device physics that took a decade of lab work to master, and first claim on the scarce substrates the process depends on, which is why the substrate supplier on the cap table matters. The spec-setter owns the clock, a component maker's play is to be standing at the seam when it strikes.

The pattern also says where to look next: by my count, of seven rack-and-board-power companies on the map, zero closed a 2026 round. Seven is a thin base, and incumbents are shipping into this transition; what is missing is startups. On a spec calendar, that reads less like an absence and more like a date that has not arrived.

Stop five: why doesn't the building have an operating system?

Follow stops three and four to their consequence. When generation, conversion, cooling, and compute are co-engineered this tightly, the facility stops being real estate that hosts machines and becomes a machine itself, and this machine has no operating system. When thousands of GPUs step through a training run in lockstep, the hall's power draw swings by tens of megawatts in seconds. Thermal headroom decides whether jobs proceed. And the incumbent software, DCIM (the data-center infrastructure management tier), was built to monitor buildings whose loads never moved. It watches; it does not model. Operators cope by over-provisioning, holding back power and cooling as safety margin because they cannot predict their own building, and that margin is capacity they bought and cannot use.

The dumbbell dates this stop precisely. Orchestration is one of the two youngest stages on my map, 54 percent of its companies founded 2024 or later. Aravolta is the shape of what is forming. A data center today runs on five separate software silos: a building-management system for HVAC, an electrical power monitoring system, SCADA for industrial controls, a network operations center, and DCIM for the IT assets. Each has its own vendor, its own dashboard, and no shared model of the building. Aravolta unifies them through what it calls a virtual PLC, a software controller that speaks to the existing hardware the way the silos' proprietary boxes do, then layers live telemetry, predictive thermal and power simulation, and anomaly detection on top, one control plane where there were five panes of glass.

Because it rides on hardware already installed, it deploys in days rather than through a retrofit, which aims it squarely at brownfield conversions: old enterprise halls being upgraded to AI densities they were never designed for, where the prediction problem is hardest and the stranded margin is largest. They are backed by YC, Susa Ventures, Pioneer Fund to name a few. Control planes are won by whoever is installed when the category consolidates. The shortening is recovered loss: megawatts that were bought, built, and then frozen as fear margin, handed back by software that makes the building predictable. The ownership is what an installed controller becomes with time: the system everything else is wired through, which is the closest software gets to being load-bearing. Windows like this do not stay open.

Stop six: where does the heat go in a state that's running out of water?

Every joule that enters the building leaves as heat, a one-megawatt hall is a one-megawatt heater. The first war over that heat, getting it off the chip, just ended. Direct-to-chip liquid cooling won, and Ecolab's $4.75 billion purchase of CoolIT is the settlement. A closed window points downstream, and downstream of the chip is the site boundary, where the tools for rejecting heat are mechanical chillers, which historically consume a fifth or more of a facility's power, or evaporative towers, which spend millions of gallons a year. The buildout concentrated in exactly the places that can least afford either. The Houston Advanced Research Center projects (primary author Dr. Margaret Cook) that Texas data-center water use will rise from about 25 billion gallons in 2025 to somewhere between 29 and 161 billion gallons a year by 2030, up to nearly three percent of state use. The width of that spread is itself a message about how little anyone knows, and water politics already shape permits.

So the open problem is precise: reject heat in dry, hot places without a compressor and without a tower's water bill. Madrone attacks it with physics that has been waiting for exactly this market. Ordinary evaporative cooling bottoms out at the wet-bulb temperature, the limit set by how much moisture the ambient air already carries. Dew-point cooling gets below that limit by sacrifice: a portion of the airstream is used to pre-cool the rest, so the working air approaches the dew point instead, colder than wet-bulb, with no compressor anywhere in the loop. The drier the climate, the bigger the gap between the two limits, which means the technique performs best precisely where the buildout went and where towers are politically hardest to permit. The boundary is real too, this is dry-air physics, built for West Texas and the desert Southwest, not for Houston's humidity, where the dew point itself sits too high at summer design conditions. Madrone packages this as external cooling units for data centers, claims roughly 30 percent less cooling power and water than a chiller-plus-tower design, carries YC backing and NVIDIA Inception membership at YC-seed scale, and is piloting in Texas. Its move is deleted steps and recovered loss at once. The compressor gone from the heat's exit path, a fifth of a facility's power handed back, and in a water-constrained county the saved water converts into the scarcest input of all, permit approval. The honest tally is that Madrone is a pure shortener, and this essay's framework does not require pretending otherwise: the last cooling war ended not in a tollbooth but in a $4.75 billion sale, and selling to the incumbent that needs your physics is a perfectly good way for a shortener's story to end. One stage back, the war ended in a $4.75 billion exit, at this stop, the entrants are seed-stage. The constraint has already walked here. The capital has not, and that is the opportunity.

Stop seven: three doors through the memory wall. Which opens first?

The walk does not end at the chip's edge; it continues at nanometer scale, and the same physics-first logic applies. Over twenty years, peak server compute grew roughly 60,000-fold while memory bandwidth grew about 100-fold and interconnect 30-fold, because compute scaled with transistor density while memory bandwidth scales with the far slower physics of pins and packaging. That gap is why the modern GPU spends much of its life waiting, why high-bandwidth memory exists, and why HBM is the cleanest physics-made chokepoint in the entire flow: three suppliers on earth, gated by packaging, sitting upstream of every accelerator, including every would-be NVIDIA killer. And the industry's own architecture choices are steepening the imbalance. The frontier has moved to mixture-of-experts models, which work by routing each token to a few specialist "experts" inside the model while the rest sit idle. That saves arithmetic (only a small fraction of the model's weights fire per token) but it saves no memory, because every expert must stay resident and reachable in case the next token needs it. So the scarce resource is increasingly memory capacity and the speed of getting to it, not arithmetic at all. (The argument is not unanimous; some recent analyses find quality-matched dense models cheaper to serve at long context. But the direction of the frontier is expert-shaped.) That seat is taken. And the software door through the wall, squeezing more from the memory you already have, closed loudly when RadixArk seeded at $400 million.

What remains are the physical doors, three of them, each held by one young company on my map, and each attacking a different layer of the same wall.

The first door is the interconnect: keep the memory where it is, but stop moving the bits over copper. Piris Labs builds a photonic interconnect over the CXL memory-pooling standard, so that GPUs can reach large pools of shared memory across light instead of wire, claiming five times lower latency and half the cost per token against the copper path. The team is the argument for taking the claim seriously. The CEO is an MIT PhD who led the optical engine on NASA's GUSTO balloon telescope (instrument-grade optics built to work where nothing can be repaired), and his co-founder ran AI infrastructure at Meta and X, the hyperscale environment the product must survive. YC-backed, NVIDIA Inception, a federal SBIR research contract, and a team that has grown from one person to seven this year.

The second door is the computation itself. If the electrical package is the ceiling, change what the package computes with. Volantis, founded by Tapa Ghosh with Roy Meade, who was the first employee at Ayar Labs and previously led HBM development at Micron, close to the exact resume this wall calls for, designs photonic chips aimed directly at the bandwidth and memory-wall limits of large-model work, moving communication inside the package onto optics. It raised a $9 million seed round with Alexandr Wang and Y Combinator founding partner Trevor Blackwell in the round, and Sam Altman an earlier backer. Photonic compute have broken hearts before, the difference in this cycle is that the memory wall, not raw FLOPS, is the binding constraint, which is the exact regime where optics' bandwidth advantage matters.

The third door leaves both the medium and the computation alone and attacks the distance instead. Chips today sit side by side on a package, so a signal passing between them travels out to the package edge and back, millimeters of wire in a world measured in nanometers. Stack the chips on top of each other and that same signal drops straight down instead, through a path hundreds of times shorter. Atum Works builds the tooling for exactly that: 3D lithography, printing structures at nanometer scale including the vertical wiring that connects stacked chips through the stack. The founding team comes out of Caltech and NASA, carries NASA awards, counts a Caltech nanoscience director and an Intel Fellow among its advisers, and holds a letter of intent with NVIDIA for co-development, which is not production but is exactly how packaging relationships with NVIDIA begin. If the memory wall is pins and packaging, Atum is the bet that the packaging itself is the thing to reinvent.

Which door opens first? I don't know, and I'd distrust anyone who claims to. Three mutually exclusive bets on the same verified imbalance; they cannot all win, and they are all the same move. It is worth saying precisely what that move shortens: a GPU stalled on memory burns full power computing nothing, so every stall is a loss, and widening the path to memory recovers it across every accelerator shipped. The ownership at this stop is unusual, because the physics IP is itself the position, with a replication half-life measured in a decade. Here the two moves collapse into a single step, which is what deep tech means.

Stop eight: what happens when compute becomes tradable before it becomes movable?

The last stop is where silicon meets software, the other heavy end of the dumbbell, and two things are breaking at once.

The first is bring-up. New silicon keeps arriving, and Etched at $10.3 billion for a transformer-only chip is just the most visible of a widening wave of accelerators, ASICs, and custom processors. But every new part pays a tax before its first production customer. Someone must rebuild the kernels, compilers, and libraries its users take for granted on NVIDIA. That is months to years of scarce specialist talent, per architecture. That tax is denominated in one of the few kinds of specialist work AI is already provably good at. Models write competent GPU kernels now, and three companies are industrializing that fact from different directions.

Infinity builds Ignition, an AI-automated kernel compiler. Point it at a model and a target (a non-NVIDIA GPU, an ASIC, an FPGA) and it generates and tunes the inference kernels that would otherwise cost a specialist team months. It raised a $15 million seed at a $100 million post from Touring Capital, has d-Matrix as an early silicon partner, and an angel list with researchers from OpenAI and Anthropic. Kernelize approaches from the compiler-infrastructure side. Its founder led AMD's work on Triton, the open kernel language the industry is standardizing on, and the company builds the hardware-agnostic stack underneath it, so that a new accelerator plugs into Triton and inherits the ecosystem instead of rebuilding it. And Wafer points the same capability backward, at the hardware already deployed. Most production PyTorch uses less than half the GPU underneath it, and the engineers who can fix that by hand command frontier-lab salaries. Wafer's agents profile live traffic, search kernel, engine, and decode configurations against it, and ship a change only when it returns identical outputs and measures faster. The loop keeps running, so the stack re-optimizes as models update and new silicon lands. The company claims two to three times better performance per dollar and says it serves over two trillion tokens a month. Every stop upstream fights for new electrons. This one gets more tokens out of the electrons already arriving. One sells the tax collapsing as a product, one builds the road that makes the tax unnecessary, and one points the machine at the installed base. All three shorten the same thing: the distance between silicon existing and silicon earning. Whether that compounds into a position depends on standardization: a bring-up tool is a vendor until the industry routes through it by default, and a compiler layer the ecosystem standardizes on is a chokepoint made of process. Triton is how that standard formed, PyTorch's own compiler now emits it by default, which quietly makes every PyTorch user a Triton user, and it is why Kernelize's founder having led it at AMD is the detail that matters.

The takeaway needs stating carefully, because it is not "NVIDIA loses." NVIDIA's real lock today is hardware and supply: the HBM it has contracted years ahead, the NVLink rack systems nobody else ships. That lock is getting stronger. What is weakening is a different thing, the software barrier that made everyone else's chips unusable and left most deployed ones running at half their capability. As that barrier falls, it cuts both ways: every new accelerator that would have needed years of kernel work becomes usable, and every GPU already installed gets cheaper to run. Both are paying customers for whoever automates the work. That is the bet here, and its appeal is that it does not require picking which chip wins. Infinity and Kernelize are paid when silicon arrives, so the wider the wave of new parts, the bigger their market. Wafer is paid whenever a workload runs, on hardware already racked. The two are complements, not halves of one market, and both grow with the buildout regardless of which chips win it.

The second is the workload itself. Every mature serving platform was shaped by chat, short stateless requests, and that tier has graduated. Baseten runs at a reported $600 million annualized revenue. But the workload is changing shape twice at once. The unit of demand is becoming a session, minutes to hours long, holding a growing memory cache hostage, alternating GPU bursts with long CPU-side orchestration. And the models themselves have gone mixture-of-experts, which turns each token into a routing problem. Only some experts fire on any given token; the hot, frequently used ones belong on expensive GPU memory, and the cold ones can live somewhere cheaper. That is an argument for spreading inference across mixed hardware rather than parking all of it on the priciest chip. Infrastructure inherits the shape of its workload, and the installed base inherited the wrong shape on both counts. Three companies are rebuilding for the new shape at three different depths.

OpenInfer rebuilds the serving layer itself. Its enterprise effort is led by Kam Eshghi, co-founder of Lightbits Labs, the NVMe-over-TCP storage company. It is building what it calls an inference OS: a control plane, Weave, that treats the session rather than the request as the unit of work, keeping session state alive across calls and splitting work across a mix of CPUs and GPUs instead of parking everything on the most expensive chip, all on an $8 million seed whose backers include Jeff Dean. Nuvacore bets the change reaches all the way into silicon. Its founder is Gerard Williams III, chief architect of Apple's M-series cores, whose last startup, Nuvia, Qualcomm bought for $1.4 billion. Sequoia seeded his new team to design CPUs from scratch for the orchestration work that gates agentic systems, on the wager that when sessions spend half their life in CPU-side coordination, the CPU stops being a commodity part. And Cedana, YC-backed, builds the missing primitive underneath everyone: GPU checkpoint and live migration. A running GPU job today is trapped on its machine. Its state cannot be frozen and relocated: gigabytes of VRAM, in-flight kernels, the collective-communication handles that bind multi-GPU jobs together. So schedulers cannot defragment clusters, spot capacity cannot be reclaimed safely, and idle hours spoil where they sit: nearly a fifth of in-execution time, in one large measurement study, and worse in the serving traces it replays. Cedana's product is the verb the fleet is missing. Pause a job, move it, resume it, the way vMotion did for virtual machines twenty years ago, a capability that quietly underwrote modern cloud economics. All three shorten, but what each could own differs, and the difference is the materials. OpenInfer's control plane and Cedana's migration primitive compound the way infrastructure software does: into dependency, the thing a fleet cannot rip out without downtime. Nuvacore's position, if it earns one, is made of harder stuff: CPU design IP, the same material its founder already turned into a moat once at Apple.

Back to the two moves

Walk back up the flow and the pattern holds at every stop. Build shortens siting, collapsing months of parcel diligence into days, and the map it accumulates while doing so is a data asset later entrants must buy or rebuild: shortening converting into ownership. Senpilot shortens the utility's own paperwork, the studies and filings where connection years are actually trapped. Soma shortens by releasing grid headroom that already exists, orchestrating generation and load from both sides of the meter, and its installed position across live assets is the part that compounds.

Apollo shortens both of nuclear's clocks, the licensing docket and the construction schedule, by shrinking the plant's largest component and keeping everything else ordinary, while Applied owns outright, holding the exclusive mPower license and wrapping a build-and-operate business around it. Voxel deletes the AC conversion stages a self-powered data center never needed: shortening at its purest. Ayr shortens the transformer wait from years to months in order to earn the qualified-vendor status that will outlast the shortage: shorten to enter, own to stay. Vertical Semiconductor shortens the last stretch of power conversion at the chip's doorstep, building to NVIDIA's published spec to get designed in.

Aravolta shortens by turning a building's stranded safety margin back into usable megawatts, on its way to owning the facility's control plane. Madrone shortens the heat's exit, deleting the chiller and most of the water bill in the geographies that can spare neither. Piris, Volantis, and Atum shorten the distance bits travel inside the package, three physical doors through a memory wall whose ownership seat, HBM, is already taken.

Infinity and Kernelize shorten silicon bring-up from engineer-years to days, and Wafer shortens the same way on hardware already racked. OpenInfer and Nuvacore shorten the waste that chat-shaped infrastructure imposes on agent sessions, one at the serving layer and one in the silicon. Cedana shortens idle hours by making a running job movable at last. Two moves, one sequence, eight stops where it is happening now.

What would make all of this wrong?

Six things, in rough order of how much they worry me.

- Political license: the fastest-moving risk on the map is not physics but permission. A statewide data-center moratorium in New York, proposed rural bans in Texas, new consumption taxes and take-or-pay tariffs from Virginia to Florida, hundreds of bills across dozens of statehouses, local groups blocking tens of billions of dollars of projects. Every stop on the walk quietly assumes the buildout keeps its social license, and that license is being renegotiated county by county right now.
- Algorithmic efficiency: quantization, distillation, and better architectures are the largest shortening lever in the whole system and deliberately outside this essay's scope; if joules per token fall an order of magnitude faster than demand grows, every bottleneck slackens at once. The historical answer is Jevons Paradox, that efficiency raises total demand, and the thing to watch is tokens-per-watt against total tokens served.
- Demand stall: American electricity demand was flat for nearly two decades before data centers reversed it, and if the AI capex cycle breaks, the bottlenecks evaporate and the chokepoints stop charging tolls.
- Interconnection reform: the generator queue finally shrank, down about a tenth from 2024 to 2025 with more than 700 gigawatts withdrawn in a single year, after years of relentless growth; if reform keeps working, the energized-land chokepoint weakens, though the load-side assumption stop one rests on still stands.
- Concentration: nearly every company in this essay ultimately sells into a demand curve set by five or six buyers, and one hyperscaler capex revision reprices the whole map at once. It is the next risk seen from the other end.
- And credit: twenty-year leases are being underwritten against three-year revenue histories, on tens of billions of data-center securitizations, and if that structured-credit assumption cracks, every queue in this essay gets shorter for the wrong reason.

What does the Spacing Guild know about chokepoints?

The Spacing Guild never mined a gram of spice. It monopolized transport, the one stage of that universe's flow nobody could route around, and converted the position into a toll on everything that moved. But the Guild's monopoly was itself hostage. Its navigators needed spice to see, so the most powerful chokepoint in the universe could be held up from further upstream, and in the end it was. Chokepoints are nested, and the winning position is the most upstream point the others cannot bypass. Enriched fuel sits upstream of the reactor license, HBM sits upstream of every accelerator, energized land sits upstream of everything. The last question to ask of any tollbooth on this flow is the one that broke the Guild: what does it, in turn, have no way around?

What I actually believe, and what I don't

The buildout will spend trillions over the next decade pushing electrons through eleven stages to turn them into words. Most of that money will chase the famous stages, the chips and the models, where the crowds already are. What the map convinced me of is that the returns sit somewhere quieter: at the stops where a young company can delete a conversion, a queue, a study, a bring-up, or a chiller, and at the handful of points, made of physics or law, where the flow simply has to pay to pass. What the map did not tell me is which specific company wins any given stop, whether the credit structure underneath the buildout holds, or how fast the efficiency lever moves. Those are open, and I've tried to mark them as open rather than dress them up.

The spice must flow, Herbert wrote, and an empire organized itself around making it true. The electrons must flow now. They are not free to travel, and that is the entire opportunity.

The Energy to Compute Flow Terminal, the 450-company map this essay is built on, is public and updated continuously. If you're building at any narrowing of this flow, or if you think any claim here is wrong, I want to hear from you. The map improves when it's argued with.